A: Passive and active walkers

Weeks 5–9

How do molecules, beads and bacteria move in water? We start from random walks and diffusion, add viscous drag and the Langevin equation, find out why life at low Reynolds number is so different from ours, and end with active particles that propel themselves and flock.

Reading

- Nelson, chapters 1–5
- Physics of active and passive motion in biology (lecture notes)
- Active matter (lecture notes)
- E. M. Purcell, Life at low Reynolds number (1977)
- T. Vicsek et al., Novel type of phase transition in a system of self-driven particles (1995)
- Background, not curriculum: Einstein 1905, Langevin 1908

Lab and project

- Lab: Brownian motion, Tuesday 9 Feb 10–14
- P1: Measuring Boltzmann's constant with a microscope, report due Sunday 7 Mar, defense Tuesday 9 Mar

Weekly exercises

	Topic	Due
E1	Orders of magnitude; Python warm-up	Wed 10 Feb
E2	Random walks	Wed 17 Feb
E3	Diffusion and tracking on synthetic data	Wed 24 Feb
E4	Drag, Reynolds number, Langevin equation	Wed 3 Mar
E5	Active Brownian particles, Vicsek model	Wed 10 Mar

Code

- `randomwalk_1d.py`, `randomwalk_exercise.py`: random-walk simulations
- `synthetic_brownian.py`: synthetic Brownian movies with known D, and a reference tracking pipeline
- `trackpy_anal.ipynb`: first steps with trackpy
- `vicsek/`: Vicsek model simulation